NATIONAL LABORATORY OF THE ROCKIES

Caution: Photovoltaic system performance predictions calculated by PVWatts® include many inherent assumptions and uncertainties and do not reflect variations between PV technologies nor site-specific characteristics except as represented by PVWatts® inputs. For example, PV modules with better performance are not differentiated within PVWatts® from lesser performing modules. Both NLR and private companies provide more sophisticated PV modeling tools (such as the System Advisor Model at //sam.nlr.gov) that allow for more precise and complex modeling of PV systems.

The expected range is based on 30 years of actual weather data at the given location and is intended to provide an indication of the variation you might see. For more information, please refer to this NLR report: The Error Report.



The energy output range is based on analysis of 30 years of historical weather data, and is intended to provide an indication of the possible interannual variability in generation for a Fixed (open rack) PV system at this location.

# RESULTS

## 611,069 kWh/Year*

*System output may range from 586,199 to 637,529 kWh per year near this location.*

| Month | Solar Radiation ( kWh / m$^2$ / day ) | AC Energy ( kWh ) |
|---|---|---|
| January | 1.82 | 23,528 |
| February | 3.29 | 37,787 |
| March | 4.29 | 52,530 |
| April | 5.04 | 58,599 |
| May | 5.99 | 69,185 |
| June | 6.04 | 66,079 |
| July | 6.86 | 75,368 |
| August | 6.40 | 71,143 |
| September | 5.52 | 61,983 |
| October | 3.62 | 43,470 |
| November | 2.15 | 26,504 |
| December | 1.92 | 24,895 |
| **Annual** | **4.41** | **611,071** |

## Location and Station Identification

| | |
|---|---|
| Requested Location | Wauconda, WA |
| Solar Resource Data Location | NSRDB Lat/Lng: 48.73, -119.02 |
| Latitude | 48.73° N |
| Longitude | 119.02° W |

## PV System Specifications

| | |
|---|---|
| DC System Size | 500 kW |
| Module Type | Standard |
| Array Type | Fixed (roof mount) |
| System Losses | 14.08% |
| Array Tilt | 25° |
| Array Azimuth | 180° |
| DC to AC Size Ratio | 1.2 |
| Inverter Efficiency | 96% |
| Ground Coverage Ratio | 0.4 |
| Albedo | *From weather file* |
| Bifacial | No (0) |

| | | | | | | |
|---|---|---|---|---|---|---|
| **Monthly Irradiance Loss** | **Jan** | **Feb** | **Mar** | **Apr** | **May** | **June** |
| | **0%** | **0%** | **0%** | **0%** | **0%** | **0%** |
| | **July** | **Aug** | **Sept** | **Oct** | **Nov** | **Dec** |
| | **0%** | **0%** | **0%** | **0%** | **0%** | **0%** |

## Performance Metrics

| | |
|---|---|
| **DC Capacity Factor** | **14.0%** |